Chao-Yuan Chiang 姜兆源 (Tera)

B.S. Student (4th year) · National Sun Yat-sen University

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Research Interests: Robot manipulation and control · System identification · Sim-to-real transfer · Learning-based policies with control

Education

National Sun Yat-sen University (NSYSU) · B.S. (in progress) — Current

Department of Mechanical and Electro-Mechanical Engineering · fourth year

Nanyang Technological University (NTU), Singapore · Exchange student — Jan. – May 2027

Courses in smart manufacturing and robotics

Experience

UROP Research Intern · City Science Lab @ Taipei Tech · National Taipei University of Technology — Jul. – Aug. 2026

- Learned robot-arm fundamentals in MuJoCo: workspace, dynamics, and impedance torque compensation.
- Tried system identification on a physical robot arm.
- Worked in a three-person team on bimanual object handover with OpenArm: VR teleoperation demos, data augmentation with Isaac Lab Mimic, SmolVLA fine-tuning, and evaluation in simulation and on the real robot.

Research & Projects

Smart Waste-Sorting Bin with Multimodal Fusion and Fallback AI Decision-Making — NSTC project: Jul. 2026 – Feb. 2027

NSTC College Student Research Project · Advisor: Assoc. Prof. Cheng-Jung Yang

Sorts waste by fusing images with impact sounds and defers uncertain items to a cloud vision-language model; with cloud review, test accuracy reaches 99.49%. **My role:** Team lead; mechatronic integration and communication among the PC, Raspberry Pi, and ESP32. Teammates led the mechanism and the core AI model.

Bimanual Object Handover on OpenArm: Real2Sim2Real Imitation Learning — Jul.–Aug. 2026

Summer UROP research internship · City Science Lab @ Taipei Tech

Trained a policy for the OpenArm bimanual robot to hand over a Pringles can: VR teleoperation demos, data augmentation with Isaac Lab Mimic, SmolVLA fine-tuning, and evaluation in simulation and on the real robot. **My role:** Member of a three-person team (summer UROP research internship).

Ball-Shooting Robot for TEL Robot Combat 2025 — Jul.–Dec. 2025

TEL Robot Combat 2025 · Advisors: Assoc. Prof. Cheng-Jung Yang and Asst. Prof. Kai-Yi Weng

A ball-shooting robot that aims automatically at a nine-hole target; our team received the Creative Technology Award (Excellence). **My role:** Team captain; hardware architecture and mechanism design of the whole robot. Teammates handled software and AI.

Robot Navigation Course Projects

Robot Navigation course, National Tsing Hua University (NTHU) AI Alliance

Implemented A* and RRT* path planning and Stanley and LQR trajectory tracking, and practiced node communication in ROS 2. **My role:** Course projects.

Publications

Modality-Robust Confidence-Gated Cascade for Edge–Cloud Waste Sorting: Design and Evaluation on a Physical Prototype — Under review

Journal manuscript · IEEE Access (Regular Manuscript)

Awards & Grants

Creative Technology Award (Excellence) · TEL Robot Combat 2025 (Tokyo Electron), 10th edition — 2025

NSTC College Student Research Project (approved) · National Science and Technology Council (NSTC), Grant No. 115-2813-C-110-129-E — 2026